HDI 2022

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Data loading and preparation

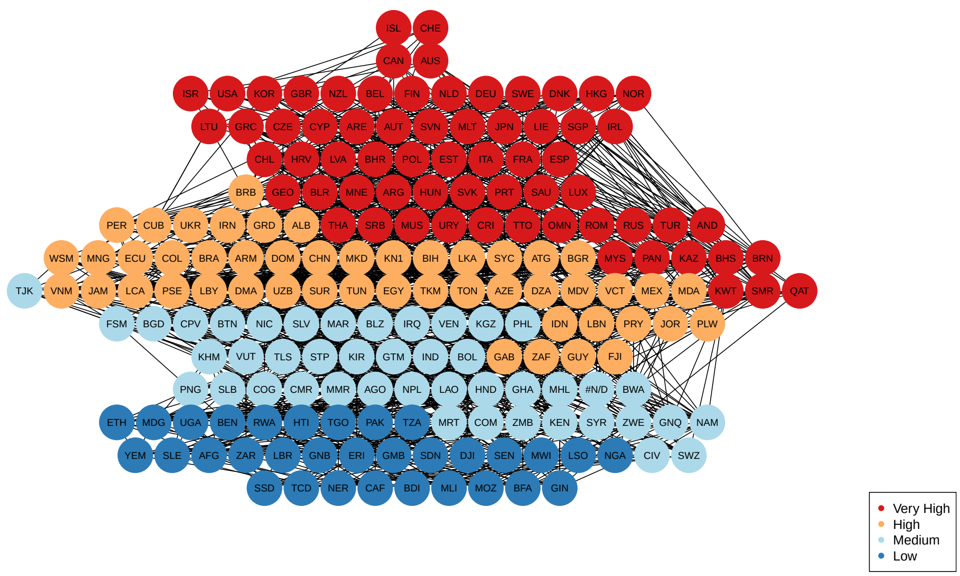
```
dataPoset <- read_delim("C:/Users/franc/OneDrive - IQS/IQS/Recerca/POSET/HDI2022.txt", del

dataPoset$Level <- ordered(dataPoset$Level, levels = c("Very High", "High", "Medium", "Low

dataPoset <- dataPoset %>% mutate(GNIpC = log(GNIpC))

X <- dataPoset$Code
dataPoset <- column_to_rownames(dataPoset, var = "Code")
dataPoset$Code <- NULL
dataPoset$Country <- NULL

dataPosetNum <- dataPoset[, c("LE", "EYS", "MYS", "GNIpC")]

head(dataPosetNum)
```

```
      LE  EYS  MYS    GNIpC
CHE 84.0 16.5 13.9 11.11145
NOR 83.2 18.2 13.0 11.07690
ISL 82.7 19.2 13.8 10.92921
HKG 85.5 17.3 12.2 11.04463
AUS 84.5 21.1 12.7 10.80442
DNK 81.4 18.7 13.0 11.00816
```

```
r <- function(x,y) all(dataPosetNum[x,] <= dataPosetNum[y,])
r <- Vectorize(r)
```

```r
Z <- outer(X, X, FUN = r)
dimnames(Z) <- list(X, X)
Z <- validate.partialorder.incidence(Z)
```

```r
palette(brewer.pal(n =4, name = "RdYlBu"))
plot(Z, col = as.factor(dataPoset$Level), pch = 16)

legend('bottomright', legend = levels(as.factor(dataPoset$Level)), col = 1:4, pch = 16)
```



Kohonen Maps

```r
dataPoset.train <- as.matrix(scale(dataPosetNum)) # make a SOM grid
set.seed(100)
dataPoset.grid <- somgrid(xdim = 4, ydim = 4, topo = "hexagonal")
```

Once the map is calculated we can look at which countries share the same cells

```r
set.seed(100)
dataPoset.model <- som(dataPoset.train, dataPoset.grid, rlen = 500, radius = 2.5, keep.dat

# str(dataPoset.model)
#data.frame(country = rownames(dataPosetNum), cell = dataPoset.model$unit.classif) %>% arr
DT::datatable(data.frame(country = rownames(dataPosetNum), cell = dataPoset.model$unit.cla
```

Show 10 entries Search:

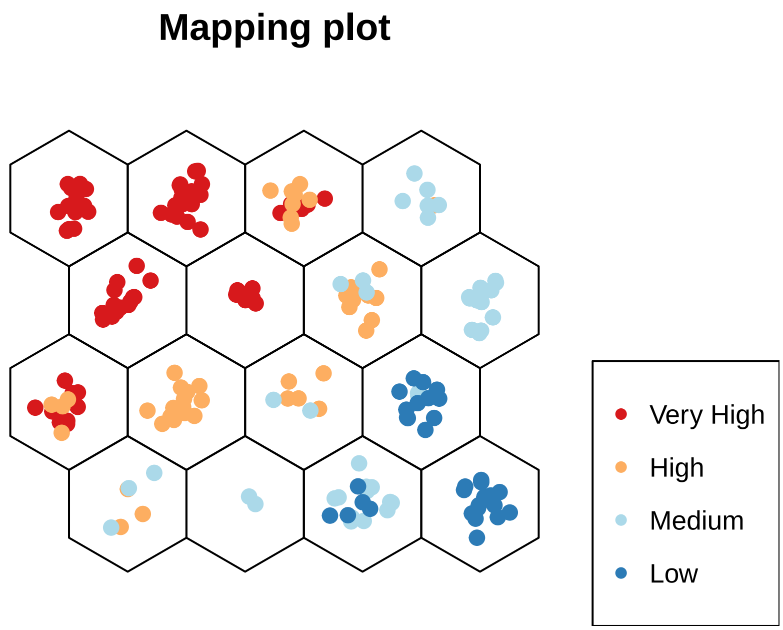
	country	cell
1	CHE	14
2	NOR	13
3	ISL	13
4	HKG	14
5	AUS	13
6	DNK	13
7	SWE	13
8	IRL	13
9	DEU	14
10	NLD	13

Showing 1 to 10 of 191 entries Previous 1 2 3 4 5 … 20 Next

```r
#plot(dataPoset.model, type = "changes")
```

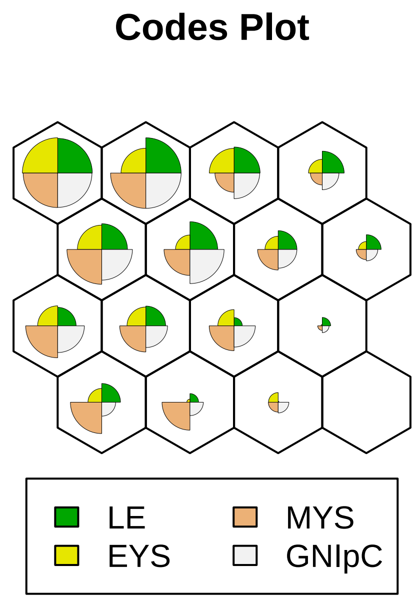
I have decided to create a 4x4 matrix to capture the different countries. The size of the map is a heuristic hyperparameter, but in any case there shouldn't be any empty cells. We observe the different countries contained in each cell, colored according to their HDI status. Cell number 1 is the one in the bottom left and cell number 16 is the one top right

```r
plot(dataPoset.model, type = "mapping", pchs = 19, shape = "straight", col = as.factor(dat
legend('bottomright', legend = levels(as.factor(dataPoset$Level)), col = 1:4, cex = 0.8, p
```



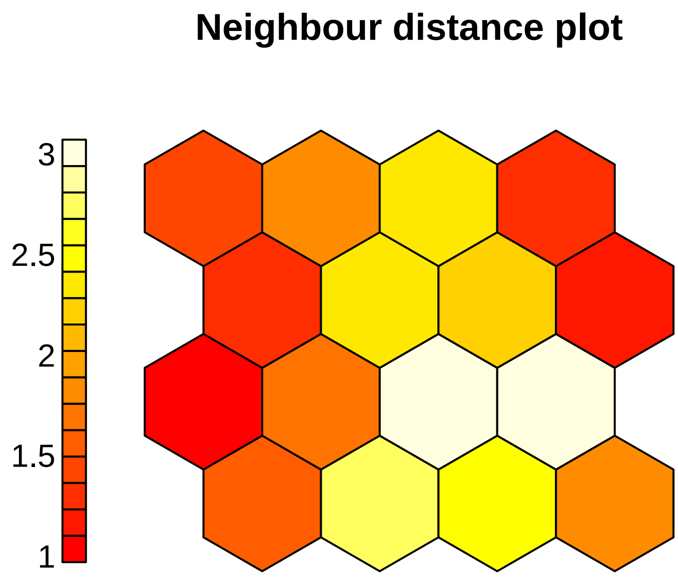
The following plot displays the characterization of each cell. Countries included in the top left cell are expected to have all variables at their maximum values

```
plot(dataPoset.model, type = "codes", main = "Codes Plot", shape="s")
```



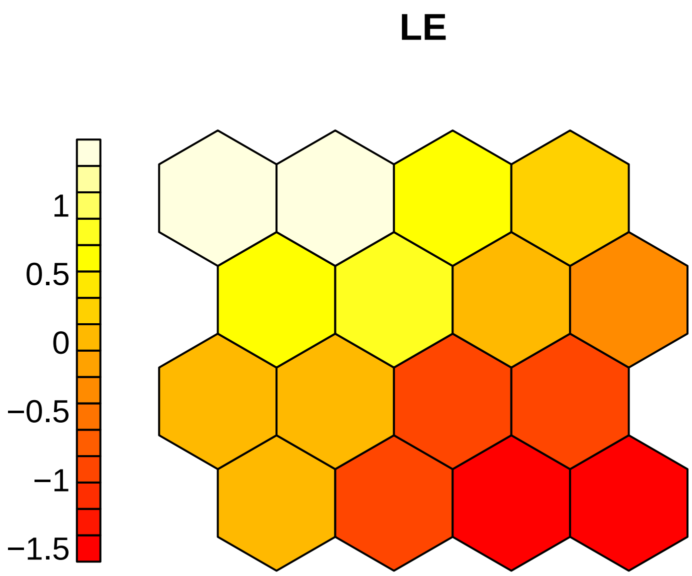
The following plot tells you how far away are each cell from their neighbours. Actually, the distance from the neighbours shows the sum of the distances to all immediate neighbours. The distance is calculated as the euclidean distance between the centers of each cell.

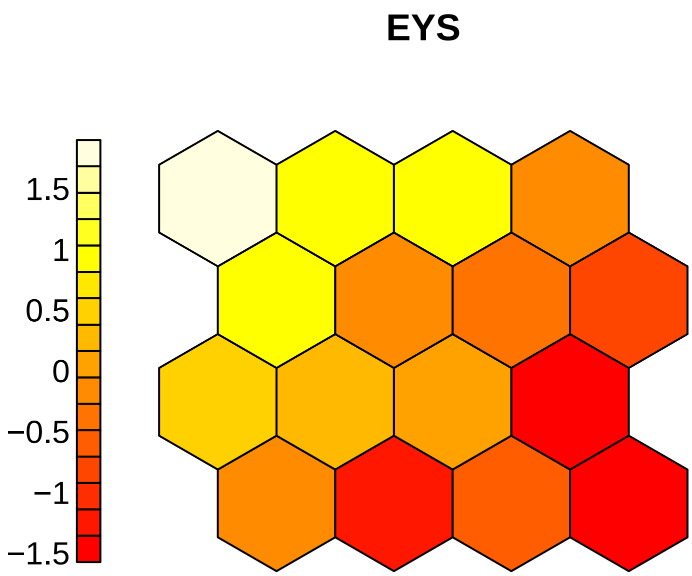
```
plot(dataPoset.model, type = "dist.neighbours", shape = "s")
```

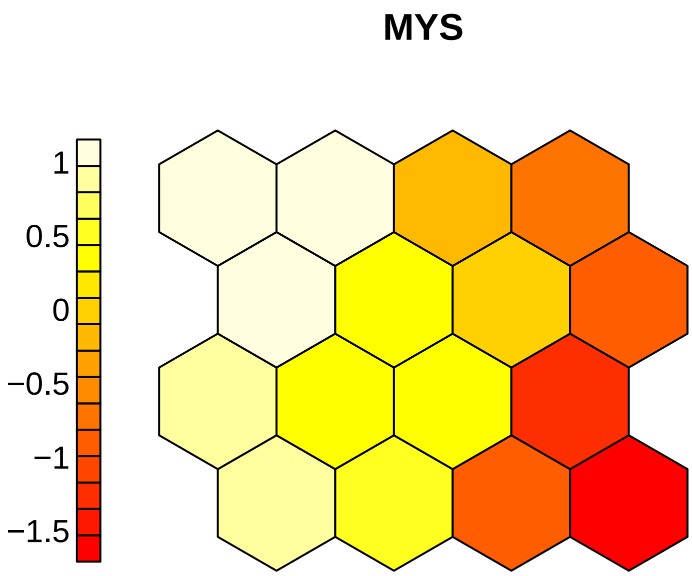


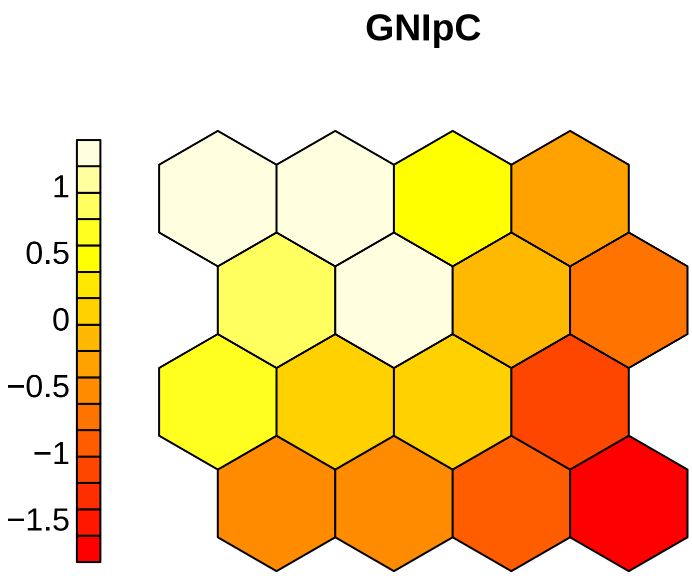
The following visualizations express the distance between cells in terms of each different variable included in the dataset.

```
heatmap.som <- function(model){
  for (i in 1:length(colnames(getCodes(model)))) {
    plot(model, shape = "straight", type = "property", property = getCodes(model)[,i],
         main = colnames(getCodes(model))[i])
    }
  }
heatmap.som(dataPoset.model)
```


LE
1
0.5
0
−0.5
−1
−1.5


EYS
1.5
1
0.5
0
−0.5
−1
−1.5

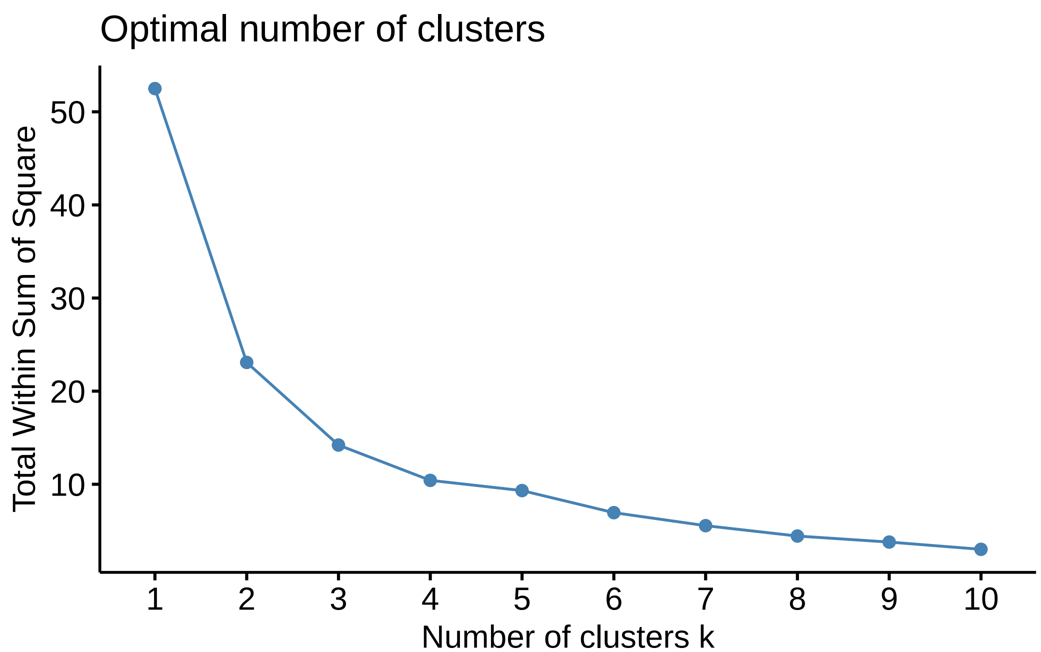

MYS
1
0.5
0
−0.5
−1
−1.5


GNIpC
1
0.5
0
−0.5
−1
−1.5

Include clusters

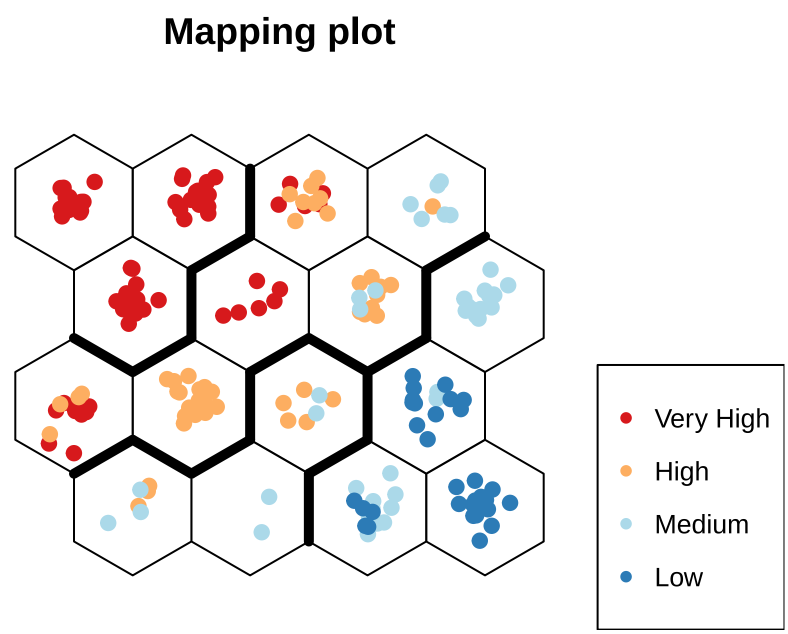
The following charts are the result of clustering the nodes of the SOM map. That is, given that each node is specified with a combination of values from each variable, one could consider which nodes are closer to which in an attempt of creating larger groups of countries in order to see if they match any known classification. We have chosen 4 clusters as it is the number of levels included in the usual HDI classifications. Besides the plot where we can choose how many clusters we want, there are two charts with the results. In the first one we see the SOM map where the different cells are separated according to the cluster. In the last chart we see the clusters with the corresponding configuration of the cells that form them.

```
set.seed(100)
fviz_nbclust(dataPoset.model$codes[[1]], kmeans, method = "wss")
```

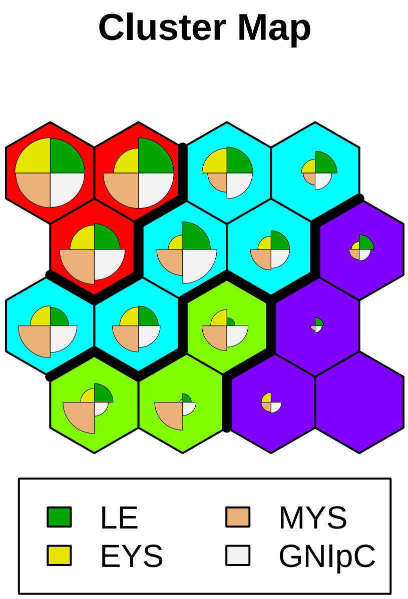



```
set.seed(100)
clust <- kmeans(dataPoset.model$codes[[1]], centers = 4) #clust <- kmeans(dataPoset.model$

dataPoset.cluster <- data.frame(dataPoset, cluster = clust$cluster[dataPoset.model$unit.cl
#tail(dataPoset.cluster, 10)
# clustering using hierarchial # cluster.som <- cutree(hclust(dist(dataPoset.model$codes[[
```

```r
#Cluster boundaries plot(dataPoset.model, type = "codes", bgcol = rainbow(1)[clust$cluster

plot(dataPoset.model, type = "mapping", pchs = 19, shape = "straight", col = as.factor(dat
legend('bottomright', legend = levels(as.factor(dataPoset$Level)), col = 1:4, cex = 0.8, p
add.cluster.boundaries(dataPoset.model, clust$cluster)
```

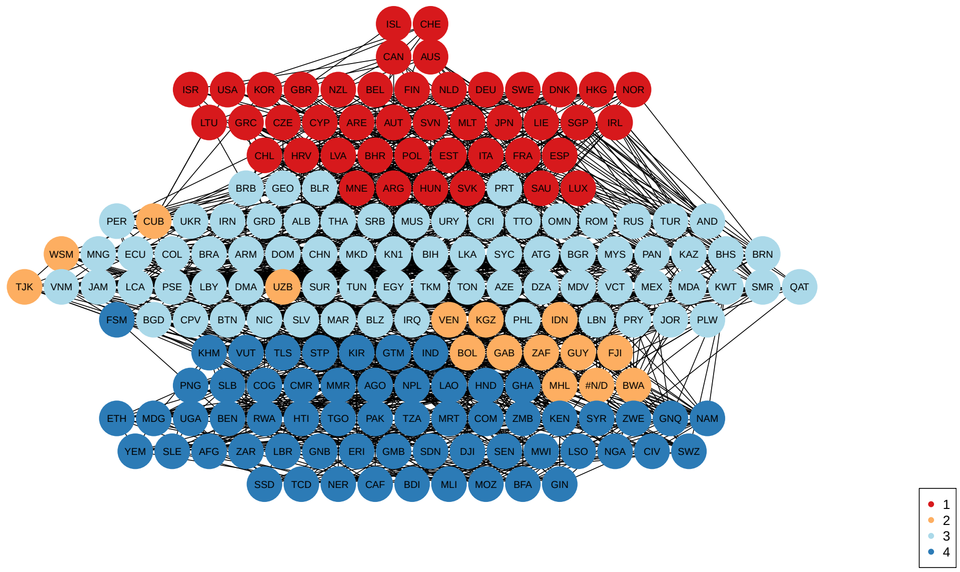



```r
plot(dataPoset.model, type = "codes", shape = "s", bgcol = rainbow(4)[clust$cluster], main
add.cluster.boundaries(dataPoset.model, clust$cluster)
```



Lastly, we are presented the same original Hasse diagram where countries are coloured according to the cluster they were assigned to

```
palette(brewer.pal(n =4, name = "RdYlBu"))
plot(Z, col = as.factor(dataPoset.cluster$cluster), pch = 16)

legend('bottomright', legend = levels(as.factor(dataPoset.cluster$cluster)), col = 1:4, pc
```



ISL
CHE
CAN
AUS
ISR
USA
KOR
GBR
NZL
BEL
FIN
NLD
DEU
SWE
DNK
HKG
NOR
LTU
GRC
CZE
CYP
ARE
AUT
SVN
MLT
JPN
LIE
SGP
IRL
CHL
HRV
LVA
BHR
POL
EST
ITA
FRA
ESP
BRB
GEO
BLR
MNE
ARG
HUN
SVK
PRT
SAU
LUX
PER
CUB
UKR
IRN
GRD
ALB
THA
SRB
MUS
URY
CRI
TTO
OMN
ROM
RUS
TUR
AND
WSM
MNG
ECU
COL
BRA
ARM
DOM
CHN
MKD
KN1
BIH
LKA
SYC
ATG
BGR
MYS
PAN
KAZ
BHS
BRN
TJK
VNM
JAM
LCA
PSE
LBY
DMA
UZB
SUR
TUN
EGY
TKM
TON
AZE
DZA
MDV
VCT
MEX
MDA
KWT
SMR
QAT
FSM
BGD
CPV
BTN
NIC
SLV
MAR
BLZ
IRQ
VEN
KGZ
PHL
IDN
LBN
PRY
JOR
PLW
KHM
VUT
TLS
STP
KIR
GTM
IND
BOL
GAB
ZAF
GUY
FJI
PNG
SLB
COG
CMR
MMR
AGO
NPL
LAO
HND
GHA
MHL
#N/D
BWA
ETH
MDG
UGA
BEN
RWA
HTI
TGO
PAK
TZA
MRT
COM
ZMB
KEN
SYR
ZWE
GNQ
NAM
YEM
SLE
AFG
ZAR
LBR
GNB
ERI
GMB
SDN
DJI
SEN
MWI
LSO
NGA
CIV
SWZ
SSD
TCD
NER
CAF
BDI
MLI
MOZ
BFA
GIN
1
2
3
4